

JavaScript Hoisting & TDZ - Quick Revision

What is Hoisting?

JavaScript behavior where **variable and function declarations** are moved to the top of their scope during compilation phase (before execution).

Key Point: JavaScript doesn't physically move code - it allocates memory during creation phase!

Variable Hoisting Summary

Declaration	Hoisted?	Initial Value	Can Access Before Declaration?
var	✓ Yes	undefined	✓ Yes (returns undefined)
let	✓ Yes	TDZ (uninitialized)	✗ No (ReferenceError)
const	✓ Yes	TDZ (uninitialized)	✗ No (ReferenceError)

var Example:

```
console.log(name); // undefined (not error!)
var name = "John";
```

let/const Example:

```
console.log(age); // ✗ ReferenceError
let age = 25;
```

Temporal Dead Zone (TDZ)

Definition: Period between entering scope and actual variable declaration where variable exists but cannot be accessed.

```
{
  // TDZ starts here for 'name'
  console.log(name); // ✗ ReferenceError
  let name = "JavaScript"; // TDZ ends here
  console.log(name); // ✓ Works!
}
```

TDZ Key Points:

- Starts at beginning of scope
- Ends when declaration is reached
- Throws ReferenceError during TDZ
- Promotes better coding practices

Function Hoisting

Type	Hoisted?	Can Call Before Declaration?
Function Declaration	✅ Completely	✅ Yes
Function Expression	✅ Variable only	❌ No

Function Declaration:

```
sayHello(); // ✅ Works!  
function sayHello() {  
  console.log("Hello!");  
}
```

Function Expression:

```
greet(); // ❌ TypeError: greet is not a function  
var greet = function() {  
  console.log("Hi!");  
};
```

Best Practices

1. Always declare variables at **top** of their scope
2. Use `let/const` instead of `var` (TDZ catches errors)
3. Declare functions before using them
4. Understand difference between function declarations vs expressions

Quick Memory Tips

- `var` : Hoisted with `undefined` - accessible but undefined
- `let/const` : Hoisted but in TDZ - not accessible until declaration
- **Function declarations**: Fully hoisted - can call anywhere in scope

- **Function expressions:** Only variable hoisted - cannot call before assignment



Common Errors

- `ReferenceError` : Accessing `let/const` in TDZ
 - `TypeError` : Calling function expression before assignment
 - `undefined` : Accessing `var` before initialization
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Remember: Hoisting exists but don't rely on it - write clear, explicit code!